Data Analytics and Automation with Alteryx & AI/ML

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Why Alteryx?

✨ **User-friendly interface**
🖱️ Provides a drag-and-drop workflow, making data analytics accessible without heavy coding.

🧹 **Data preparation & blending**
🔄 Easily cleans, transforms, and combines data from multiple sources.

📊 **Advanced analytics**
🤖 Supports predictive modeling, statistical analysis, and machine learning techniques,
💡 helping businesses make data-driven decisions efficiently.

🔗 **Integration with other tools / Workflow automation**
⚙️ Automates repetitive data processes, saving time, manual effort, and improving efficiency.
📈 Integrates with tools like Tableau, Power BI, and various databases to enhance visualization and reporting capabilities.

Top Organizations Leveraging Alteryx

SIEMENS energy

JPMorganChase

Goldman Sachs

Walmart

THE HOME DEPOT

Deloitte.

EY

KPMG

pwc

What are we going to learn?

🗄️ **SQL (Structured Query Language)**
Learn how to store, retrieve, and manage data efficiently.

🔄 **ETL (Extract, Transform, Load)**
Understand how to collect data from multiple sources, clean it, and load it for analysis.

📈 **Data Analytics**
Explore techniques to analyze data and uncover meaningful insights for decision-making.

⚙️ **Workflow Creation**
Design and build structured data workflows for smooth and scalable processes.

📊 **Data Visualization**
Present data visually using charts, graphs, and dashboards for better understanding.

📝 **Reporting**
Create clear and effective reports to communicate insights and findings.

🤖 **Automation**
Automate repetitive data tasks to improve efficiency and reduce manual effort.

🧠 **AI & ML (Artificial Intelligence & Machine Learning)**
Gain foundational knowledge of intelligent systems that learn from data and make predictions.

INSTALLATION

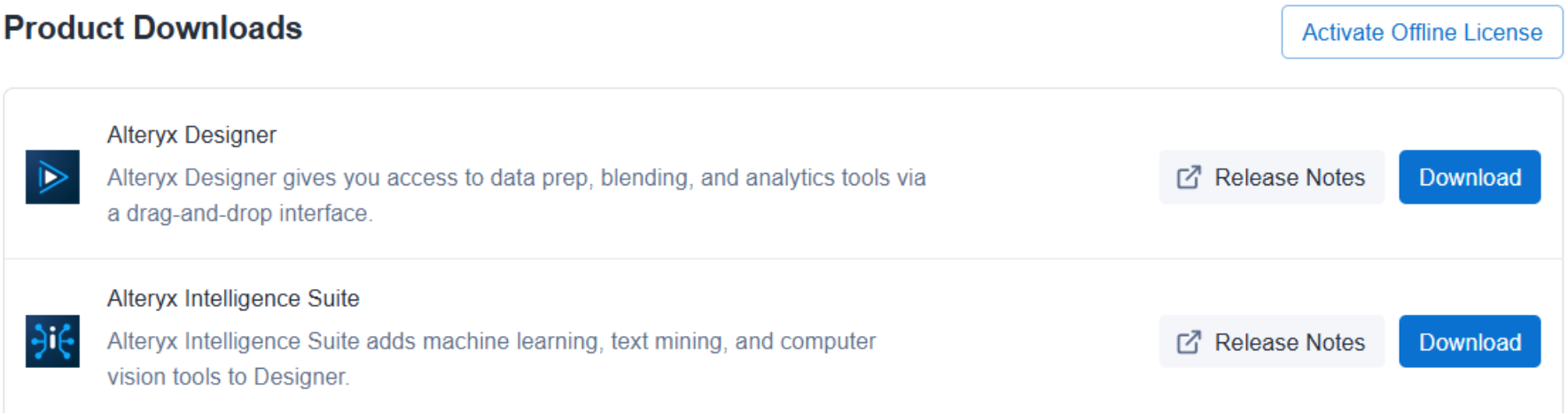
👉 Get student license from SparkED program: https://www.alteryx.com/sparked/learning-programs/students

✔️ Complete the registration process and wait for the verification to complete.

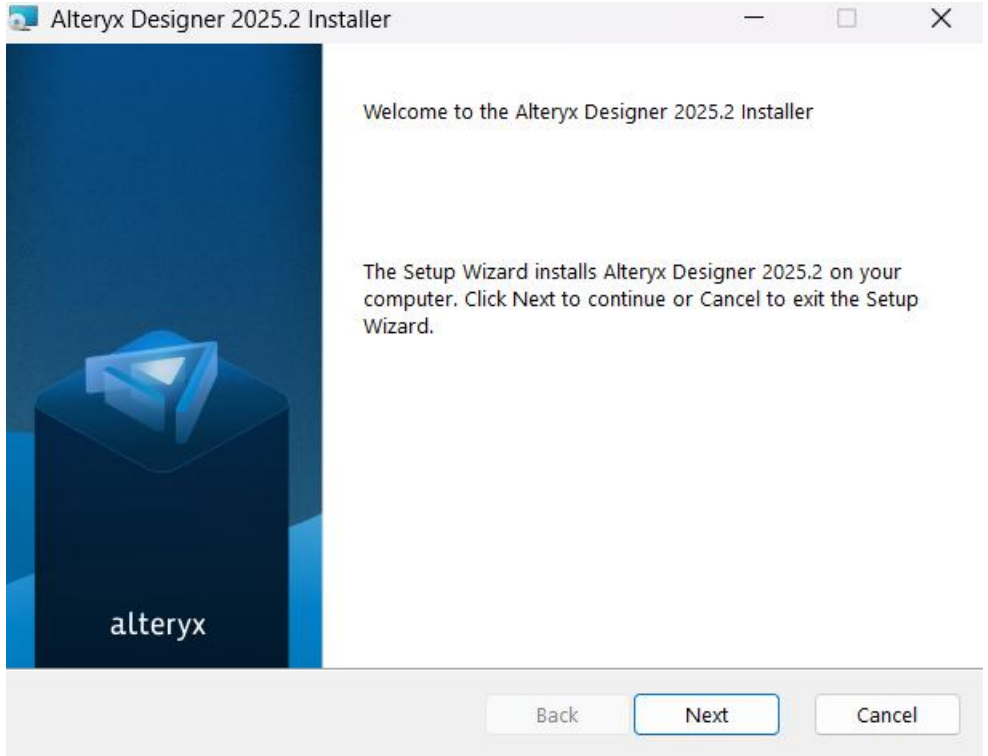
✅ Once successful, the Alteryx license will be delivered to your email.

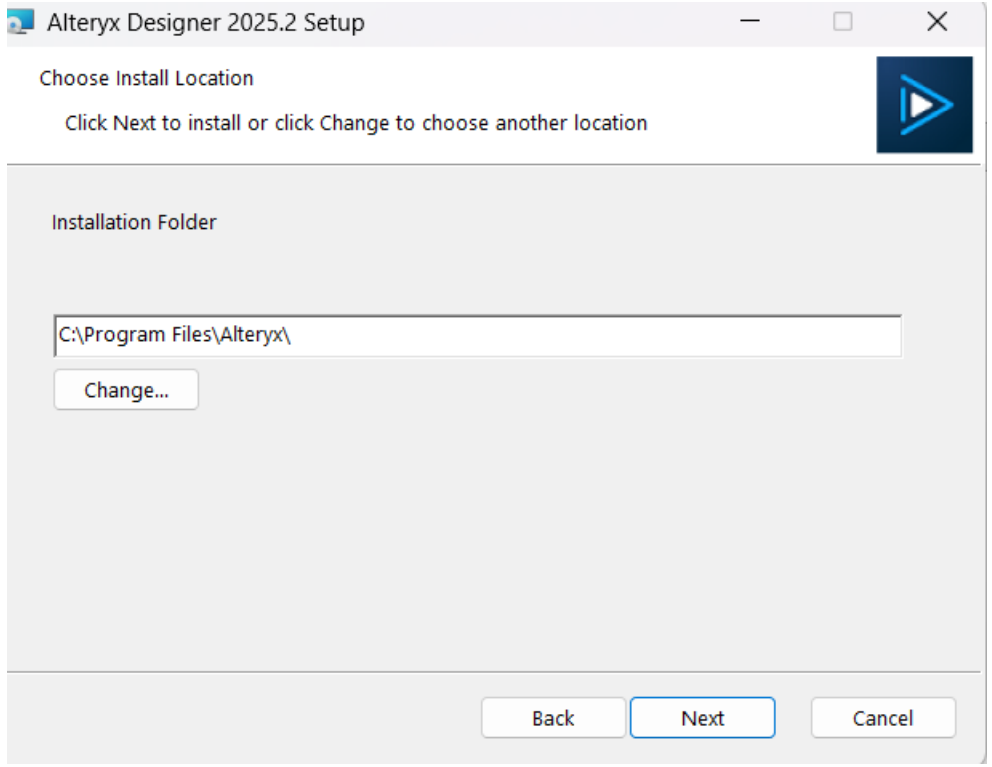
🌐 Visit: https://us1.alteryxcloud.com/license-portal/

🔐 Login and download the setup files

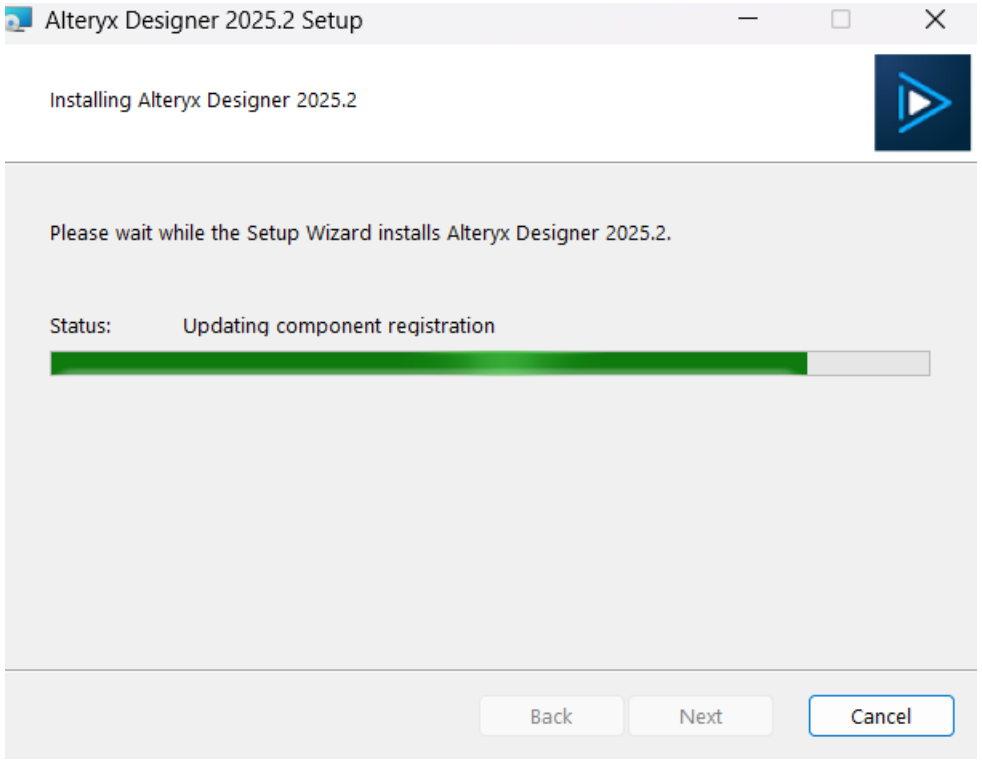


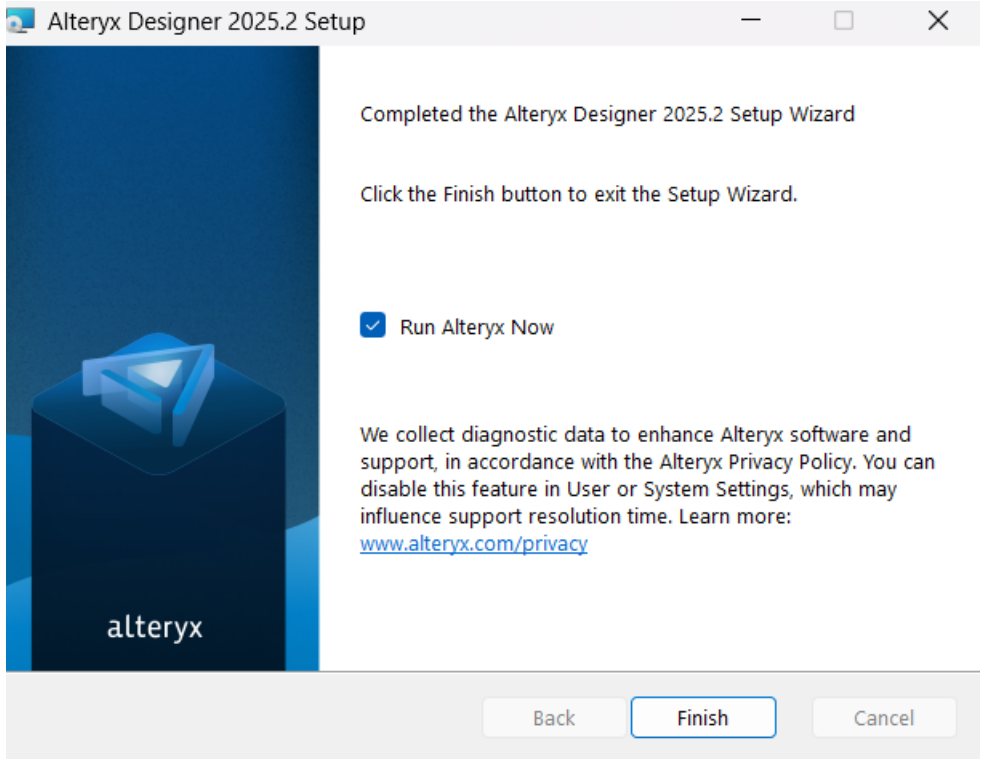
⚙️ Execute the installer in your desktop



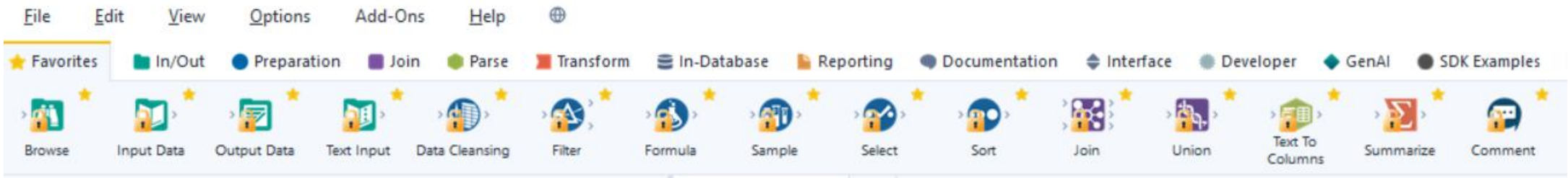


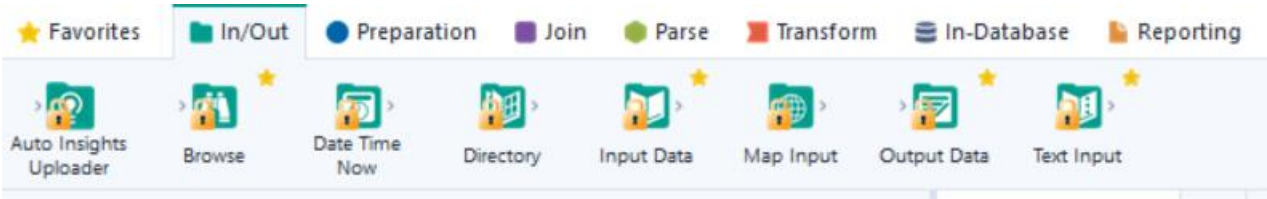
🛠️ Click ‘Next’ and complete the steps, until you get the complete screen and hit ‘Finish’.

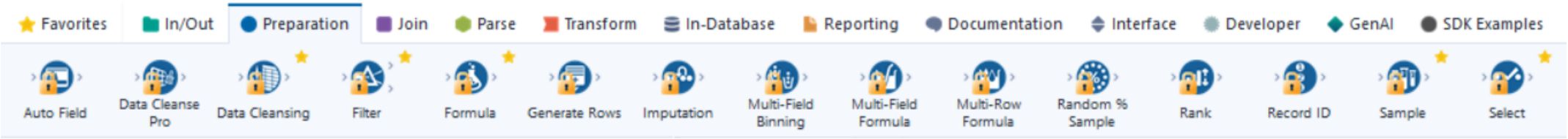




BEFORE ACTIVATION





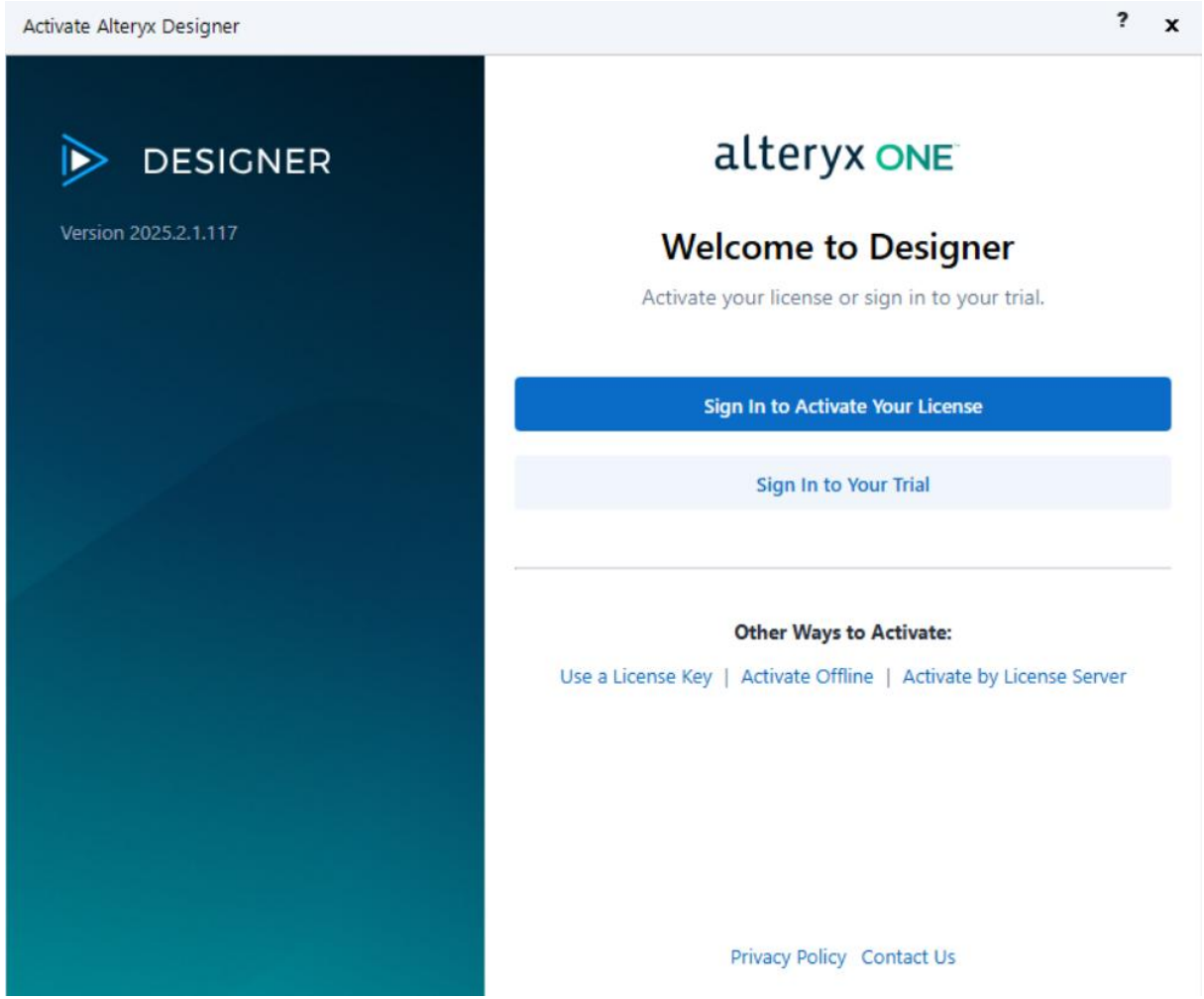


ACTIVATION

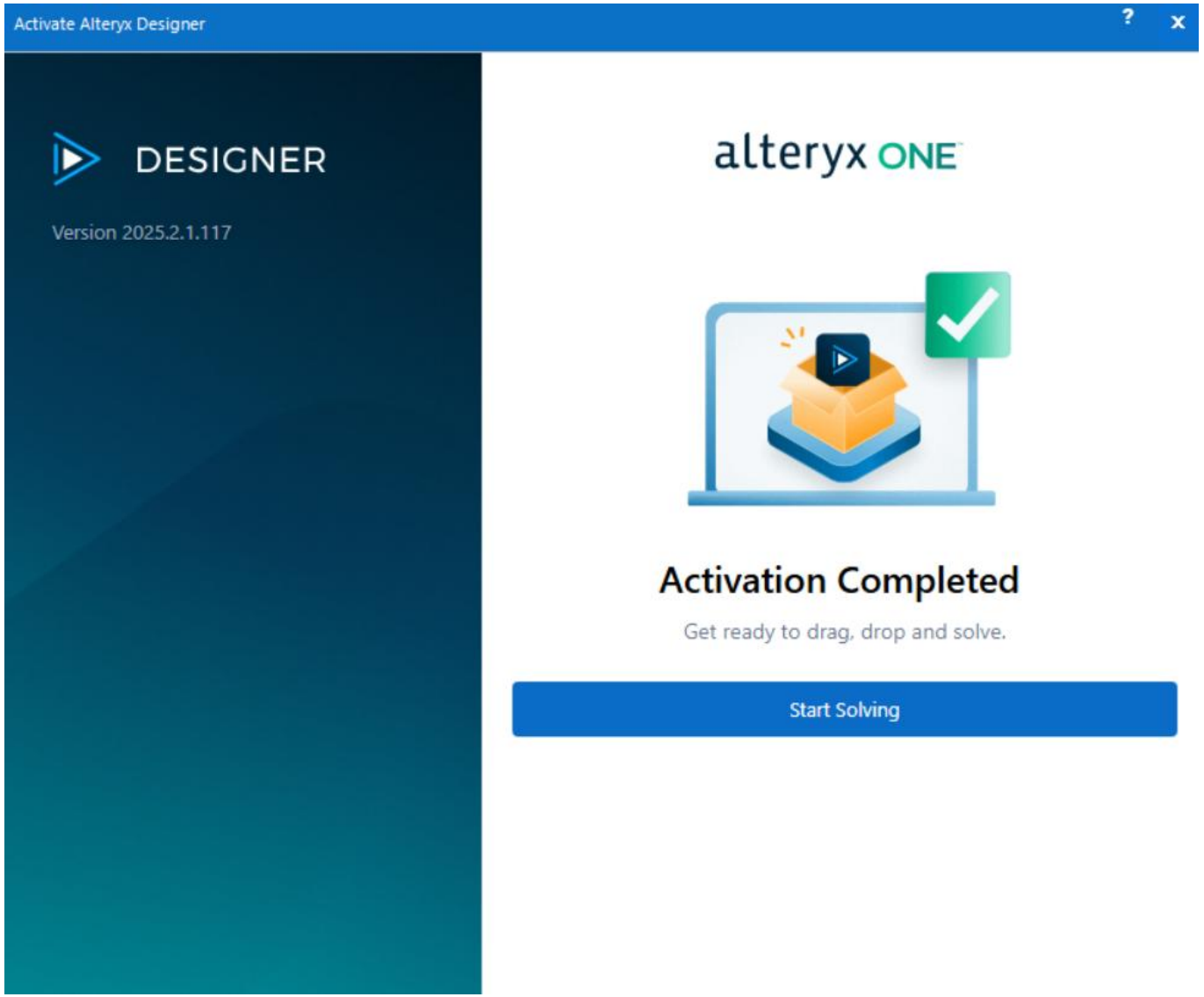
Click on “Use a License Key”.

Enter the registered email address.

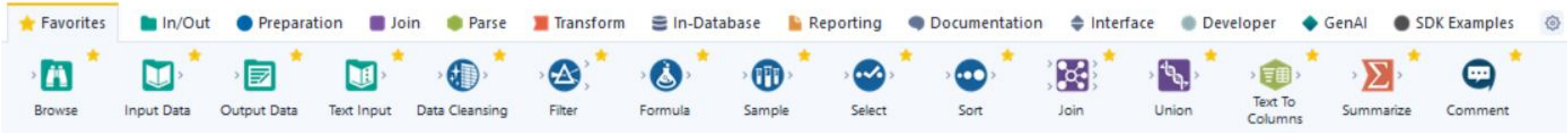
Enter the received product key from email.



Upon successful activation, will be able to view the “Activation Completed” screen.



AFTER ACTIVATION



Let’s Begin Alteryx Learning

Alteryx Interface Overview

Core Areas

🧩 **Canvas (Workflow Area)**
Build workflows by dragging and connecting tools
🛠️ **Tool Palette**
Access tools organized by categories
🔍 **Search Bar**
Quickly locate tools
⚙️ **Configuration Window**
Customize selected tool settings

Execution & Navigation

📊 **Results Window**
View output data, logs, and errors
▶️ **Toolbar**
Run, save, and manage workflows
🔗 **Workflow Connections**
Define data flow between tools
🧭 **Overview Window**
Navigate large workflows easily

🧠 **Workflow Properties**
Set global workflow configurations
🎛️ **Interface Designer**
Create apps and macros

Alteryx File Types & Documentation

Core File Types

- 📄 **.yxmd – Workflow File**
Standard file to create and run workflows
- 🔁 **.yxmc – Macro**
Reusable workflow components
- 🧩 **.yxwz – Analytic App**
Interactive workflows with user inputs
- 📦 **.yxzp – Packaged Workflow**
Workflow bundled with all dependencies

Data & Integration Files

- 🗄️ **.yxdb – Alteryx Database**
High-performance data storage format
- 📊 **.hyper – Tableau File**
Integration with Tableau
- 📑 **.csv / .xlsx / .txt**
Common external data formats

🔗 **Documentation:** https://help.alteryx.com/

Demo – Alteryx Interface

Some Important Alteryx Designer Tools

Input/Output Tools

Browse
Text Input
Input Data
Output Data

Preparation Tools

Select
Filter
Formula
Unique

Join and Combine Tools

Join
Union
Find Replace
Append Fields

Parse Tools

DateTime
Regex
Text To Columns
XML Parse

Transform Tools

Cross Tab
Transpose
Count Records
Summarize

Machine Learning Tools

Auto ML
Classification
Fit
Predict

🚨 **Note:** Not all tools are covered here.

🔗 More tools and Info: https://help.alteryx.com/current/en/designer/tools/designer-tools-list.html

ETL

📥 Extract → 🛠️ Transform → 📤 Load

📥 **Extract**: Get data from different places (like apps, files, databases)

🛠️ **Transform**: Clean it, fix it, and make it useful

📤 **Load**: Put it into a system (like a data warehouse) for use

✨ Think of it like cooking:
🥕 Get ingredients → 🍳 Cook & prepare → 🍽️ Serve the dish

Demo – How to use Alteryx Tools

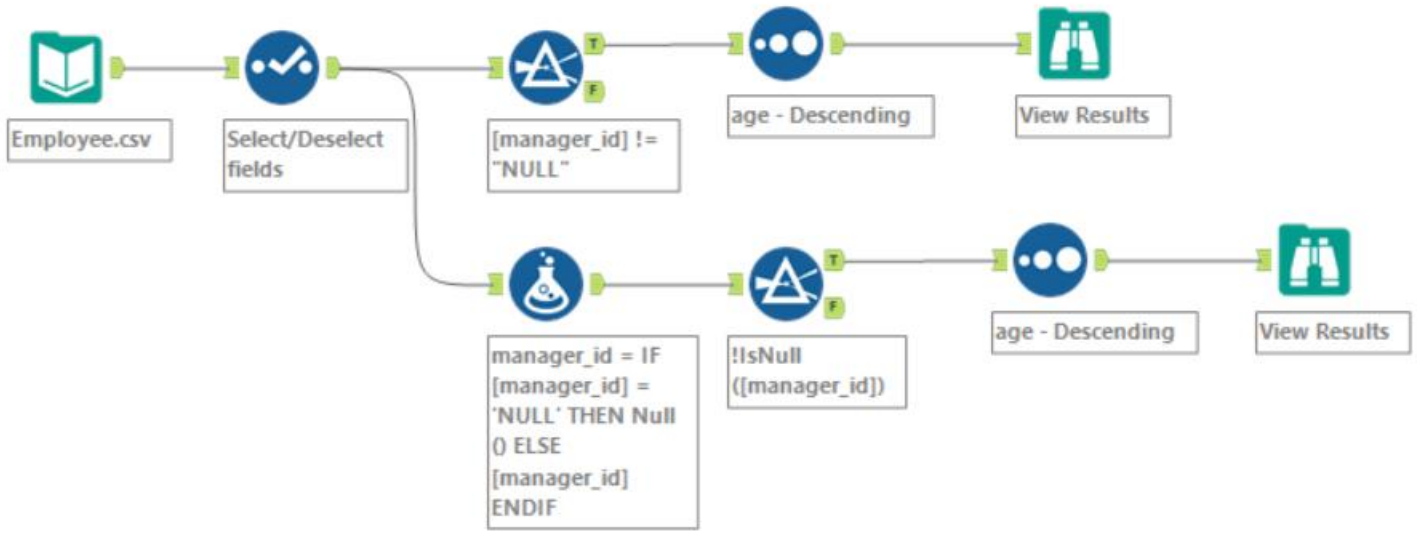
SQL with Alteryx

F J O W G H S D O L
(*From Join On Where Group Having Select Distinct Order Limit*)

SQL Order of Execution
1. **FROM** – Identify source tables
2. **JOIN** – Combine tables (if any)
3. **ON** – Specify join conditions
4. **WHERE** – Filter rows before grouping
5. **GROUP BY** – Group rows
6. **HAVING** – Filter groups
7. **SELECT** – Choose columns / expressions
8. **DISTINCT** – Remove duplicates
9. **ORDER BY** – Sort results
10. **LIMIT / OFFSET** – Restrict number of rows returned

Demo 1: SQL with Alteryx

Alteryx Workflow:



SQL:

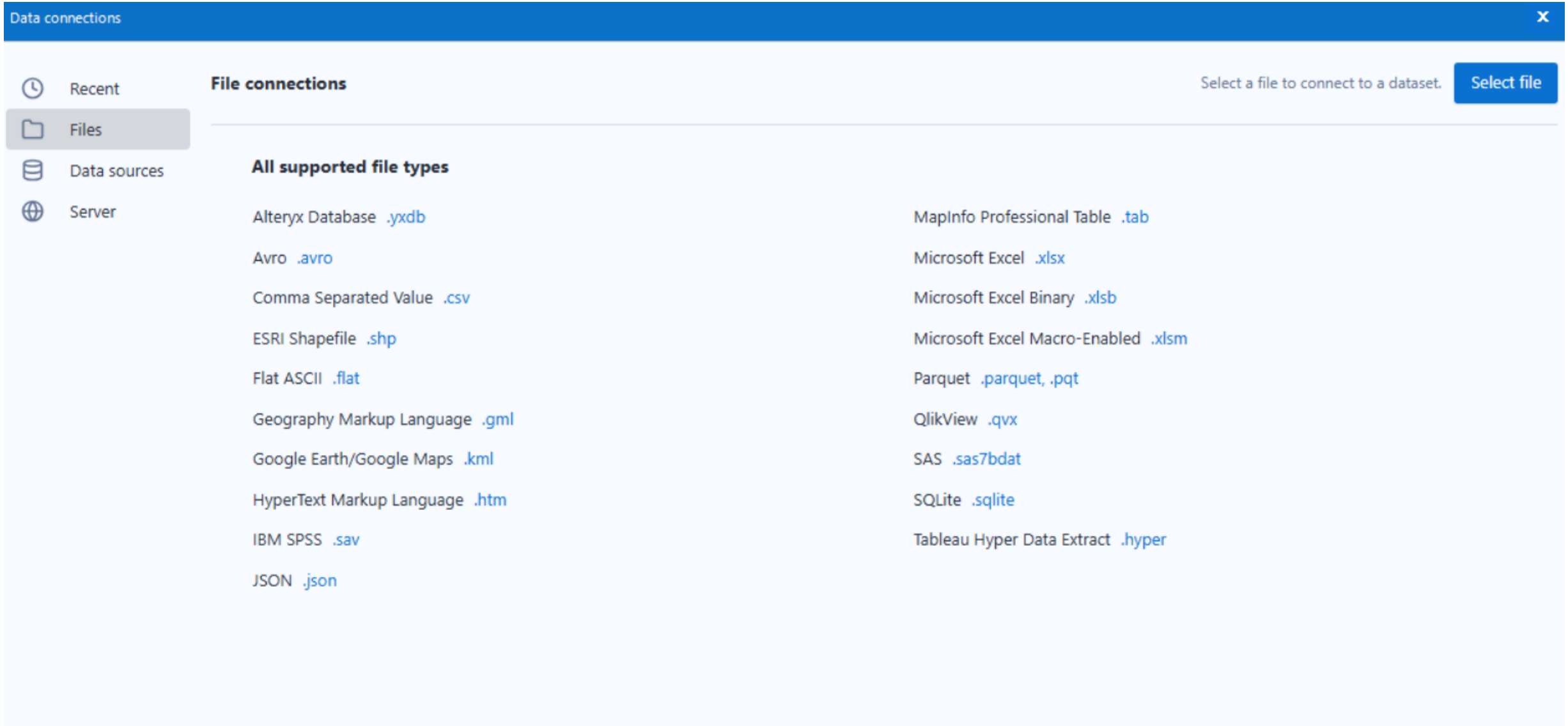
```
SELECT * FROM EMPLOYEE
WHERE MANAGER_ID IS NOT NULL
ORDER BY AGE DESC;
```

Link to Demo Workflow

💡 **Learning from this Demo:** Know the difference between String ‘NULL’ value and Database ‘NULL’.

🚨 **Note:** This is for understanding purposes only.

Alteryx Supported File Types



Workshop - 1

📥 Load **GoldRate** and **ExchangeRate** datasets using pipe (|) delimiter.

🏷️ Use **first row as header** for column names

🔗 Join both datasets on **CurrencyCode**

🚫 Remove duplicate **CurrencyCode** column after join

🔄 Reorder columns and assign correct **data types** (Date, Decimal, etc.)

💱 Convert gold rates to INR

➤ Create new column: **GoldPrice = GoldRate × CurrencyValue**

🎯 Select required fields: **Date, Country, GoldPrice**

📊 Generate Output 1: **Average GoldPrice per Country**

📅 Generate Output 2: **Filter GoldPrice for a specific Date per Country**

📦 [Link to Workshop Workflow]

Demo 2: Data into Actionable Insights

What do we do here?

Streamline and refine customer data to generate accurate, actionable results.

📥 **Input Customer Data** – Import and organize incoming data

✅ **Validate Data** – Ensure accuracy and consistency

🔍 **Filter Data** – Remove irrelevant or unwanted entries

🔄 **Transform Data** – Convert data into the required format

🧩 **Consolidate Data** – Combine data from multiple sources

📤 **Deliver Results** – Generate and present the final output

📦 Link to Demo Workflow

Workshop - 2

📂 Load Student, Department, Subject, and Marks datasets

🔗 Join all tables using StudentID, DeptID, and SubjectID

🧹 Clean and select required fields (Student, Department, Marks)

🧮 Summarize marks to calculate TotalMarks per Student

📊 Group data by Department and Student for aggregation

🏆 Rank students within each Department by TotalMarks (descending)

🥇 Select top student per Department and generate final output table

📦 Link to Workshop Workflow

Demo 3: From Refined Data to Visual Insights

✨ **Data Processing & Insight Generation Workflow**

📥 **Ingest Core Data** – Load primary datasets into the system

⚙️ **Execute Data Operations** – Apply essential processing steps

📌 **Integrate Lookup Data** – Enrich data with additional reference inputs

🔄 **Transform Data** – Restructure and standardize for usability

📊 **Aggregate Data** – Summarize data to extract meaningful patterns

📈 **Visualize & Report** – Create impactful dashboards and reports

📦 Link to Demo Workflow

Workshop - 3

🚲 Load Stations, Customers & Rentals data

🔗 Join all tables (Stations + Customers + Rentals)

⏱️ Parse StartTime and EndTime using DateTime tools

🧮 Calculate Ride Duration using DateTimeDiff formula

⚠️ Filter rides with duration > 60 minutes (Long Duration)

🌙 Extract hour from StartTime to find peak usage

📊 Summarize total rides by hour for peak analysis

🏆 Summarize total spending per customer to find top customers

🚉 Summarize station usage for busiest start/end stations

💰 Calculate total revenue by Bike Type and Membership Type

📈 Create charts for rides, revenue, and Station Usage

📑 Build final reporting layout (tables + visuals)

📤 Export final dashboard as PDF/Excel report

📦 [Link to Workshop Workflow]

Alteryx Resources

- Designer
- Workflows
- Apps and Macros
- Tools
- Functions
- Data Sources
- File Types Support
- Connection Setup

Exercises

📦 Link to Exercises

Task Time

📦 Link to Task Datasets

Machine Learning & Intelligence Suite

Demo 4 – Computer Vision

Document Intelligence using PDF

📦 Link to Demo Workflow

Demo 5 – Text Mining

What we'll cover

🧹 Text Pre-Processing

😊 Sentiment Analysis

🧠 Topic Modeling

☁️ Word Cloud

📦 Link to Demo Workflow

Demo 6 – Machine Learning

Implement and Compare the Model Results

📦 Link to Demo Workflow

Exercises

📦 Link to Exercises

Exploring other tools and features

Alteryx Resources on AI/ML

- Machine Learning
- Computer Vision
- Text Mining
- GenAI

Q & A and Discussions

Unlock Your Certification Journey

🔗 Certification and Prep Guide:
https://community.alteryx.com/t5/Certification-Exams/bd-p/product-certification#designer-desktop

Let’s Stay in Touch

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Thank You!